Erasure-Aware Byzantine-Robust Aggregation for Hierarchical UAV-Swarm Federated Learning: When Link Telemetry Hurts

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Abstract—Federated learning (FL) lets a UAV swarm train a shared model without moving raw sensor data, but tactical links lose packets and some UAVs may be compromised. Byzantine-robust aggregation rules usually assume that every update arrives intact, while channel-aware FL rarely assumes adversaries. We study a hierarchical UAV-swarm FL system whose uplinks are driven by a measured air-to-ground received-power trace, and a trusted-reference reputation aggregator with three levels of channel awareness: none, packet-erasure masking, and full link telemetry (packet delivery ratio, reliability, geometry, mobility and handover) inside the reputation update. In a prespecified paired study of 720 runs (10 seeds, three model-poisoning attacks, six aggregators), erasure masking raises accuracy under attack at a 5 dB link by 4.2 pp (95% CI [2.2, 5.9]), whereas adding telemetry changes it by −3.5 pp ([−5.1, −2.1]), because it rejects 41.2% of honest updates. We trace this to an asymmetric reputation rule whose stationary point drifts toward a UAV's worst rounds when its link is weak.

Index Terms—Federated learning, UAV swarm, Byzantine robustness, packet erasure, hierarchical aggregation, reputation.

I. Introduction

UAV swarms increasingly train perception models collaboratively, and FL [1] keeps each UAV's imagery on board while exchanging only model updates [2]. Two impairments act on those updates at once. First, air-to-ground links fade, hand over between base stations (BSs) and miss deadlines, so updates arrive with missing packets [3]. Second, captured or spoofed UAVs can send poisoned updates. Robust rules such as Krum [4], coordinate-wise median [5] and FLTrust [6] address the second problem but treat every received vector as complete; channel-aware FL and anti-jamming designs [7] address the first but do not assume adversarial clients. Reputation-based defenses are attractive for persistent swarm members, yet they can themselves be gamed [8].

A natural idea is to let the BS's own link telemetry modulate trust: a UAV on a poor link should not be blamed for an update that looks unusual because of the channel. Our earlier single-seed experiments suggested such a context-calibrated rule beat a cosine-weighted baseline taken from prior UAV-FL work [9]. This letter tests that idea under controlled conditions and finds that it does not hold. The main contributions of this letter are as follows:

1) We model hierarchical UAV-swarm FL whose uplink packet erasures are driven by a measured DeepSense 6G UAV received-power trace [10], with BS association, handover and model-poisoning attackers.
2) We propose an erasure-aware reputation aggregator and isolate its two channel-aware mechanisms, erasure masking and telemetry-weighted reputation, in a prespecified, paired 720-run simulation study.
3) We derive the steady-state reputation of the update rule and show that it explains why telemetry lowers accuracy as the link weakens, while erasure masking raises it.

The rest of this letter is arranged as follows. Section II describes the system model and Section III the proposed aggregation. Section IV analyses its reputation update at steady state. Section V presents simulation results, and Section VI concludes the letter.

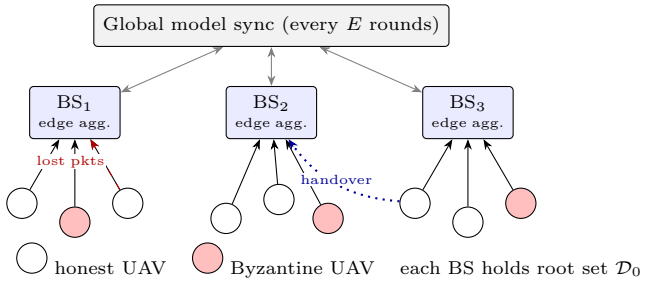


Fig. 1. Hierarchical UAV-swarm FL: each BS aggregates the (partially received) updates of the UAVs it serves; BS models are synchronized every E rounds.

II. System Model

A. Hierarchical learning

K UAVs train a model $\mathbf{w} \in \mathbb{R}^P$ and are served by B BSs (Fig. 1). In round t, UAV k attached to BS b starts from the edge model $\mathbf{w}_b^t$, runs local SGD on its private data $\mathcal{D}_k$ and uploads $\boldsymbol{\Delta}_k^t$. BS b aggregates the updates it receives into $\mathbf{w}_b^{t+1} = \mathbf{w}_b^t + \mathrm{Agg}(\cdot)$. Every E rounds the BS models are averaged, weighted by each cell's number of training samples, and broadcast. Each BS also holds a

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Code, data and the scripts that produce every figure in this letter: https://github.com/sasank-555/uav-swarm-fl.

small trusted root set $\mathcal{D}_0$, disjoint from all UAV data, on which it computes a reference update $\mathbf{r}_b^t$ each round, as in FLTrust [6].

B. Erasure channel

An update is split into packets of L parameters. The uplink SNR γ_k^t of UAV k is obtained by replaying a measured UAV received-power sequence [10] around an assumed reference SNR $\bar{\gamma}$ and adjusting it for the 3-D distance to each virtual BS. Each packet is lost independently with probability

$$p_k^t = \big(1 + \exp\big(a(\gamma_k^t - \gamma_{\text{th}})\big)\big)^{-1}, \tag{1}$$

and an update that misses the round deadline is dropped. The BS therefore observes $\tilde{\boldsymbol{\Delta}}_k = \mathbf{m}_k \odot \boldsymbol{\Delta}_k$, where $\mathbf{m}_k \in \{0,1\}^P$ marks delivered coordinates. Association uses a hysteresis-and-time-to-trigger handover rule on received power.

C. Threat model

A fixed set of f UAVs is Byzantine. They follow the protocol but replace their update by a sign-flip ($-\lambda\boldsymbol{\Delta}_k$), an ALIE-style update [11] ($\boldsymbol{\mu} - z\boldsymbol{\sigma} \odot \operatorname{sign}(\boldsymbol{\mu})$, from the honest updates of the same cell), or an inner-product-manipulation (IPM) update [12] ($-\epsilon\boldsymbol{\mu}$). The ALIE and IPM attackers observe honest updates in their cell, a deliberately strong assumption. Telemetry is measured by the BS and is assumed authentic.

III. Proposed Erasure-Aware Reputation Aggregation

Let $\mathcal{S}(\mathbf{x};\mathbf{m})$ restrict $\mathbf{x}$ to the coordinates where $\mathbf{m} = 1$. Each BS runs the following steps (index t dropped).

1) Clipping. With ρ_k the RMS of the delivered entries of $\tilde{\boldsymbol{\Delta}}_k$, scale the update by $s_k = \min(1, \operatorname{med}_j \rho_j/\rho_k)$.

2) Masked similarity. Compare each update with the root update on delivered coordinates only, so lost packets do not make an honest update look dissimilar: $c_k = \cos\big(\mathcal{S}(s_k\tilde{\boldsymbol{\Delta}}_k;\mathbf{m}_k), \mathcal{S}(\mathbf{r};\mathbf{m}_k)\big)$.

3) Behavior score. $b_k = \operatorname{clip}\big(c_k/\operatorname{med}_{j:c_j>0} c_j,\, 0,\, 1\big)$.

4) Reputation. With memory β,

$$T_k \leftarrow T_k + \eta_k\,(b_k - T_k), \qquad \eta_k = \begin{cases} (1-\beta)\,e_k, & b_k > T_k, \\ 1-\beta, & b_k \le T_k, \end{cases} \tag{2}$$

so that bad behavior is penalized at full rate and a weak link cannot shield an attacker that drops its own packets. The evidence $e_k \in [0,1]$ slows rewards on poor links: $e_k = \text{PDR}_k \cdot R_k \cdot h_k \cdot g_k \cdot v_k$, the product of the packet delivery ratio, long-term reliability, a handover penalty ($h_k = 0.5$ during a handover), a distance factor and a speed factor.

5) Masked weighted average. UAVs with $T_k < \tau$ get zero weight; otherwise $\omega_k = n_k T_k$, with $n_k = |\mathcal{D}_k|$ and a staleness discount. Each coordinate is averaged only over the UAVs that delivered it:

$$\text{Agg}_p = \frac{\sum_k \omega_k\, m_{k,p}\, s_k \tilde{\Delta}_{k,p}}{\sum_k \omega_k\, m_{k,p}}. \tag{3}$$

Variants. CCRA (full) uses all steps with telemetry evidence e_k. Mask-only sets $e_k = 1$ but keeps the masks. No-context additionally sets $\mathbf{m}_k = \mathbf{1}$, so lost packets enter steps 1–5 as zeros, which is how a channel-unaware server sees them. (The SINR calibration of c_k in our code is inactive here because the study has no residual parameter noise, so the full-versus-mask-only contrast isolates e_k.)

IV. Steady-State Analysis of the Reputation

Consider one UAV whose evidence is constant, $e_k = e \in (0,1]$, and whose behavior scores $b \in [0,1]$ are i.i.d. with $\mathbb{E}[b] > 0$. By (2), the expected reputation step from level T is $(1-\beta)\,g_e(T)$ with

$$g_e(T) = e\,\mathbb{E}\big[(b-T)^+\big] - \mathbb{E}\big[(T-b)^+\big]. \tag{4}$$

Proposition 1: g_e has a unique root $T^\star(e) \in (0,1)$, the steady state of the mean reputation dynamics. $T^\star(1) = \mathbb{E}[b]$, $T^\star(e)$ is increasing in e, and $T^\star(e) \to \operatorname{ess\,inf} b$ as $e \to 0$.

Proof: g_e is continuous and strictly decreasing in T, with $g_e(0) = e\,\mathbb{E}[b] > 0$ and $g_e(1) = -\mathbb{E}[1-b] < 0$, so the root exists and is unique. For $e = 1$, $g_1(T) = \mathbb{E}[b] - T$. Since $\partial g_e/\partial e = \mathbb{E}[(b-T)^+] \ge 0$ while g_e decreases in T, the root rises with e. As $e \to 0$, $g_e(T) \to -\mathbb{E}[(T-b)^+]$, which is negative for every $T > \operatorname{ess\,inf} b$. ∎

The asymmetry intended to stop packet-dropping attackers therefore has a side effect. An honest UAV with non-IID data has noisy scores, and on a weak link (small e) its reputation settles near its worst rounds instead of its average, so it falls below the gate τ more often. An attacker's scores are already near zero, so a smaller e adds little suppression. Section V tests this prediction.

V. Simulation Results

A. Simulation Setup

Table I lists the values assumed for the parameters. Attackers are chosen per seed and kept fixed across methods. The channel is either ideal or the same measured trace replayed at $\bar{\gamma} = 15\,\text{dB}$ (mean PDR 99.8%) and at $\bar{\gamma} = 5\,\text{dB}$ (mean PDR 65.0%). The baselines are FedAvg [1], coordinate-wise median [5] and FLTrust [6], the last given the same root set and a comparable local-step budget.

The full factorial design crosses six aggregators, four attack settings (including clean), three channels and 10 paired seeds. A run that diverges numerically is scored at chance (10%), never discarded. The primary contrast, CCRA minus mask-only at 5 dB minus the same at 15 dB, and its 1 pp decision threshold were fixed before the runs. Intervals are 95% percentile bootstraps over seeds (10,000 resamples). All other contrasts are exploratory and not adjusted for multiple comparisons.

B. Robustness and Stability

Fig. 2 and Table II show that FedAvg and the coordinate-wise median diverged in 41 and 26 of their

TABLE I
Values Assumed for the Parameters

Parameter	Value
UAVs K; BSs B; attackers f; sync E	10; 3; 3; 5
Rounds; local training	100; one SGD epoch
Model	CNN, 156k parameters
Data	aerial CIFAR-10, Dirichlet(0.5) [13]
Root set $\lvert\mathcal{D}_0\rvert$; β; τ	500 images; 0.5; 0.3
Packet size L; a; γ_{th}	1024; 1; 3 dB
Reference SNR $\bar{\gamma}$	ideal, 15 dB, 5 dB
Sign flip λ; ALIE z; IPM ϵ	5; 1.5; 0.5
Seeds; runs	10; 720

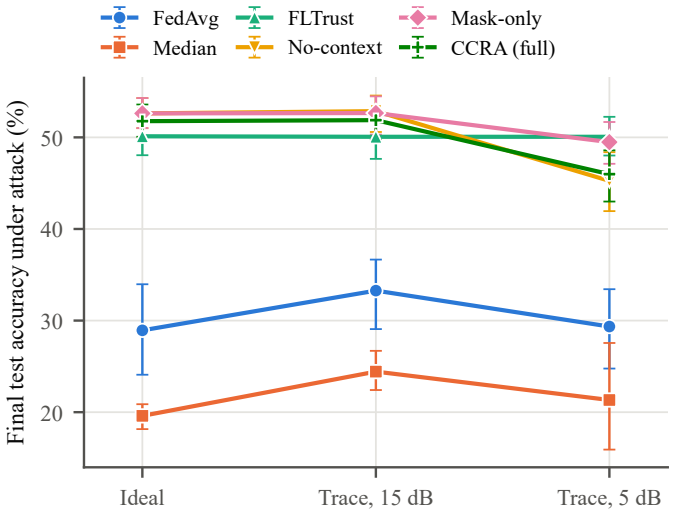


Fig. 2. Final accuracy under attack (mean of sign-flip, ALIE and IPM over 10 seeds; bars: 95% bootstrap CI). Diverged runs count as 10%.

120 runs respectively, whereas no root-referenced method (FLTrust and all three CCRA variants) diverged. The median is likely fragile here because each cell holds only a few UAVs, so a single attacker in a cell can already move the median. Across all runs, mask-only attains the highest accuracy (52.8% versus 37.3% for FedAvg). Against FLTrust it gains 2.6 pp ([0.9, 4.5]) under attack at 15 dB, but at 5 dB the two are indistinguishable (−0.6 pp, [−2.0, 1.0]; exploratory).

C. Which Channel Awareness Helps

Fig. 3 separates the two mechanisms. On the ideal channel the masks are all ones, so mask-only and no-context coincide. As delivery degrades, erasure masking becomes valuable: at 5 dB it adds 4.2 pp ([2.2, 5.9]). Telemetry does the opposite. At 5 dB, full CCRA minus mask-only is −3.5 pp ([−5.1, −2.1]; 46.0% versus 49.5%), and the prespecified interaction lies entirely below −1 pp. Combining the two mechanisms leaves an inconclusive net effect relative to no-context.

TABLE II
Run outcomes over all 720 runs (120 per method)

Method	Diverged	All runs (%)	Attacked, 5 dB (%)
FedAvg	41/120	37.3	29.4
Median	26/120	29.3	21.3
FLTrust	0/120	50.6	50.1
No-context	0/120	51.4	45.3
Mask-only	0/120	52.8	49.5
CCRA (full)	0/120	51.1	46.0

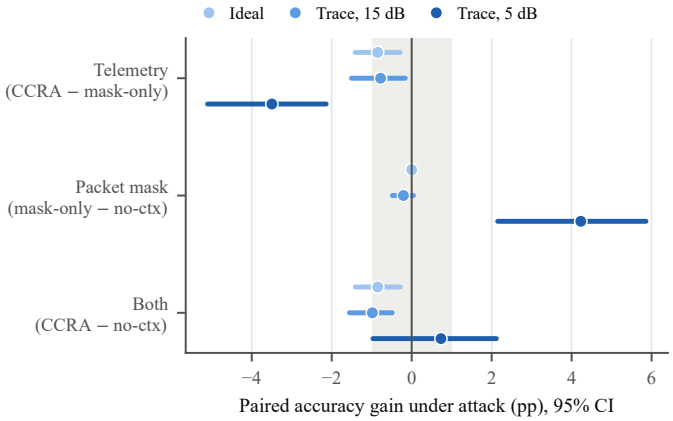


Fig. 3. Paired ablation contrasts under attack. The gray band marks the prespecified ±1 pp region of practical equivalence.

D. Validation of the Steady-State Analysis

Fig. 4 matches Proposition 1. With telemetry, the gate rejects 41.2% of honest updates versus 13.8% for mask-only, and admits only slightly fewer attacker updates (8.8% versus 13.5%). The rule trades a small gain in attacker filtering for a large loss of honest data, and the loss grows as the link weakens, where honest data is already scarce. FLTrust, which has no memory, rejects only 4.8% of honest updates, consistent with its resilience at 5 dB. These rates are descriptive and support, but do not prove, the causal account.

E. Limitations

The trace comes from one measured BS with virtual neighbors, the SNR anchors are assumptions rather than calibrated values, the run includes few handovers, and we use one dataset and model. Telemetry spoofing and adaptive attacks on the reputation itself [8] are not tested.

VI. Conclusion

In a hierarchical UAV-swarm FL system with measured-trace packet erasures, the channel-aware component that pays off is simple: compare and average each update only on the coordinates that actually arrived. Feeding link telemetry into an asymmetric reputation update looked reasonable but measurably hurt accuracy on weak links by suppressing honest UAVs. Future work will redesign the reward rule so that weak links do not bias reputation downward, evaluate it on fresh seeds and higher-mobility

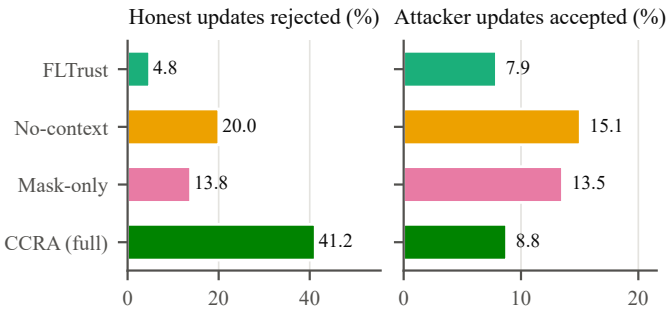


Fig. 4. Reputation-gate decisions under attack at 5 dB, averaged over attacks and seeds (percent of received updates).

traces, and study adaptive and telemetry-spoofing attackers.

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